Team Relay

Sprint #3 Review

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Sprint Goals



- HIPAA Training
- CRUD for organization
- Licensing
- Kit ID Generation

Sprint #3 tasks

~~FGS-23~~	Unique KitID	scurti03
~~FGS-39~~	Implement sealing of key file for transfer of p...	scurti03
~~FGS-44~~	Implement API for encrypting and decryptin...	scurti03
~~FGS-47~~	Tests for KitID	yivanova
~~FGS-40~~	Implement hybrid wrap of patient data using ...	scurti03
~~FGS-33~~	HIPAA Training	yivanova
~~FGS-41~~	HIPAA Training	yuoo
~~FGS-42~~	HIPAA Training	dshin17
~~FGS-34~~	Implement Organization model	dshin17
~~FGS-35~~	Implement CRUD API endpoints	dshin17
~~FGS-36~~	URL routing in backend	dshin17
~~FGS-38~~	Find schema for Central	dshin17
~~FGS-48~~	Implement sealing of key file for transfer of p...	scurti03
~~FGS-46~~	Refactor current KitID to mint once, ensure n...	yuoo
~~FGS-55~~	Add meeting notes to confluence, meeting tr...	yivanova

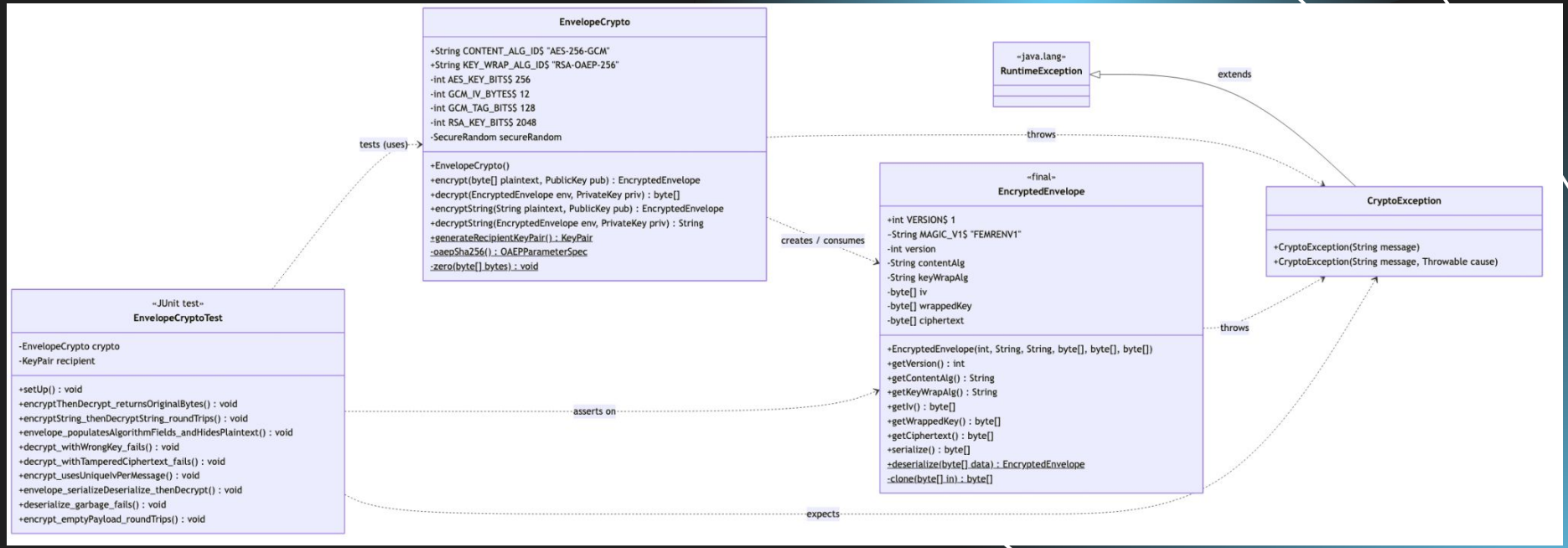
Backlog

FGS-56	Add git smart commits	To Do	3	
FGS-45	Implement licensing handshake between user admin and kit.	To Do	3	S
FGS-57	Setup JIRA time tacking	To Do	3	
FGS-49	Adding current documentation and diagrams to confluence, update a...	In Progress	1	S
FGS-58	Implement different languages	To Do	5	
FGS-54	HIPPA Training	In Progress	1	S
FGS-51	Read documentation on confluence pertaining to patient data transfe...	In Progress	1	S
FGS-60	Make a matrix with what we're going to support + future upgrades	To Do	1	
FGS-61	Plan kitID migration	To Do	3	Y

Progress for fEMR Clients

- Defined the empty Organization Model
- Resolved dependency compatibility issues to match requirements
- Made migration 0029 for adding clients to databases
- Implemented client_view and single_client_view

Encryption


EnvelopeCrypto
+String CONTENT_ALG_ID$ "AES-256-GCM"
+String KEY_WRAP_ALG_ID$ "RSA-OAEP-256"
-int AES_KEY_BIT$$ 256
-int GCM_IV_BYTE$$ 12
-int GCM_TAG_BIT$$ 128
-int RSA_KEY_BIT$$ 2048
-SecureRandom secureRandom
+EnvelopeCrypto()
+encrypt(byte[] plaintext, PublicKey pub) : EncryptedEnvelope
+decrypt(EncryptedEnvelope env, PrivateKey priv) : byte[]
+encryptString(String plaintext, PublicKey pub) : EncryptedEnvelope
+decryptString(EncryptedEnvelope env, PrivateKey priv) : String
+generateRecipientKeyPair() : KeyPair
-oaepSha256() : OAEPParameterSpec
-zero(byte[] bytes) : void
«java.lang»
RuntimeException
extends
throws
tests (uses)
creates / consumes
«final»
EncryptedEnvelope
+int VERSION$ 1
-String MAGIC_V1$ "FEMRENV1"
-int version
-String contentAlg
-String keyWrapAlg
-byte[] iv
-byte[] wrappedKey
-byte[] ciphertext
+EncryptedEnvelope(int, String, String, byte[], byte[], byte[])
+getVersion() : int
+getContentAlg() : String
+getKeyWrapAlg() : String
+getIv() : byte[]
+getWrappedKey() : byte[]
+getCiphertext() : byte[]
+serialize() : byte[]
+deserialize(byte[] data) : EncryptedEnvelope
-clone(byte[] in) : byte[]
CryptoException
+CryptoException(String message)
+CryptoException(String message, Throwable cause)
throws
«JUnit test»
EnvelopeCryptoTest
-EnvelopeCrypto crypto
-KeyPair recipient
+setUp() : void
+encryptThenDecrypt_returnsOriginalBytes() : void
+encryptString_thenDecryptString_roundTrips() : void
+envelope_populatesAlgorithmFields_andHidesPlaintext() : void
+decrypt_withWrongKey_fails() : void
+decrypt_withTamperedCiphertext_fails() : void
+encrypt_usesUniqueIvPerMessage() : void
+envelope_serializeDeserialize_thenDecrypt() : void
+deserialize_garbage_fails() : void
+encrypt_emptyPayload_roundTrips() : void
asserts on
expects

Kit ID Implementation Progress

Completed

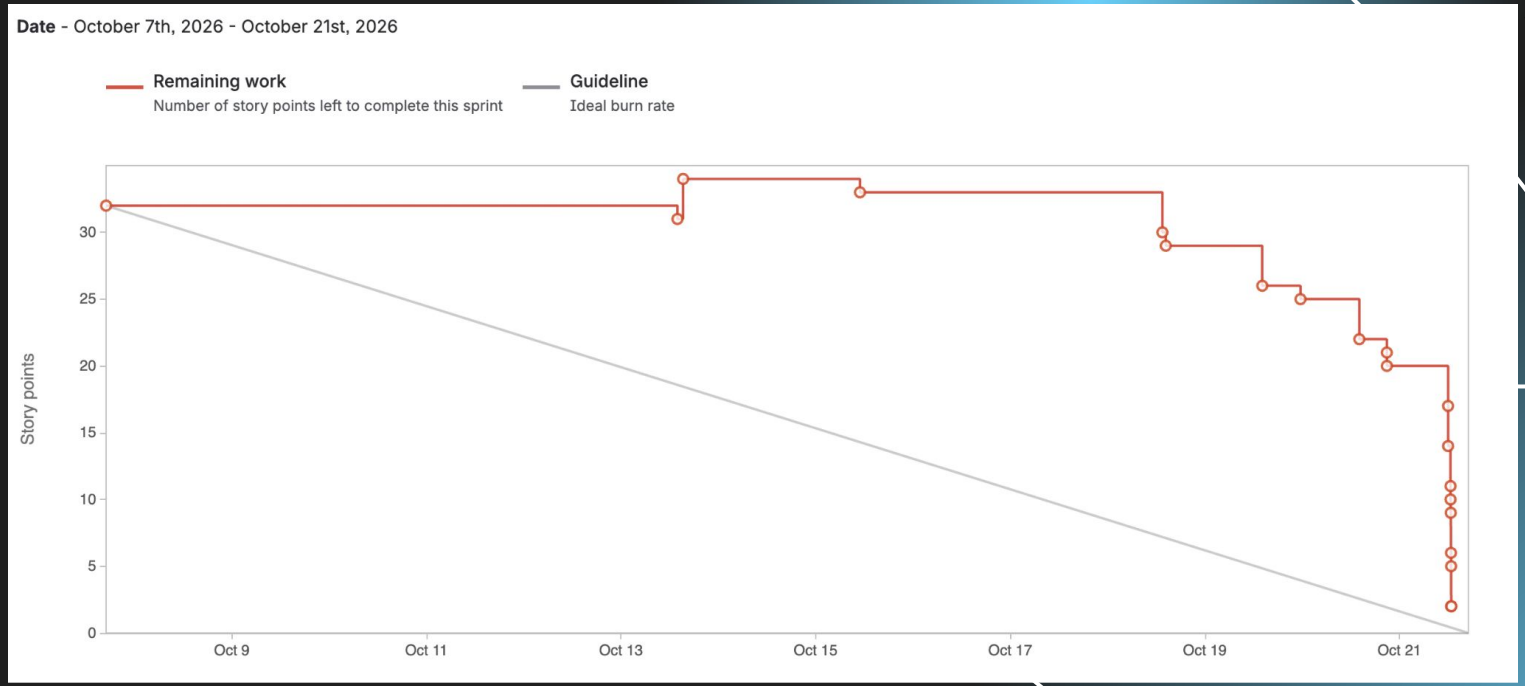
- Implemented UUID-based Kit ID generation
- Added persistent local database storage
- Added UUID validation and duplicate prevention
- Added support for legacy ID tracking
- Implemented identity service and repository
- Added tests for kitID
- Started migration planning for existing kitIDs

Kit ID Implementation Progress

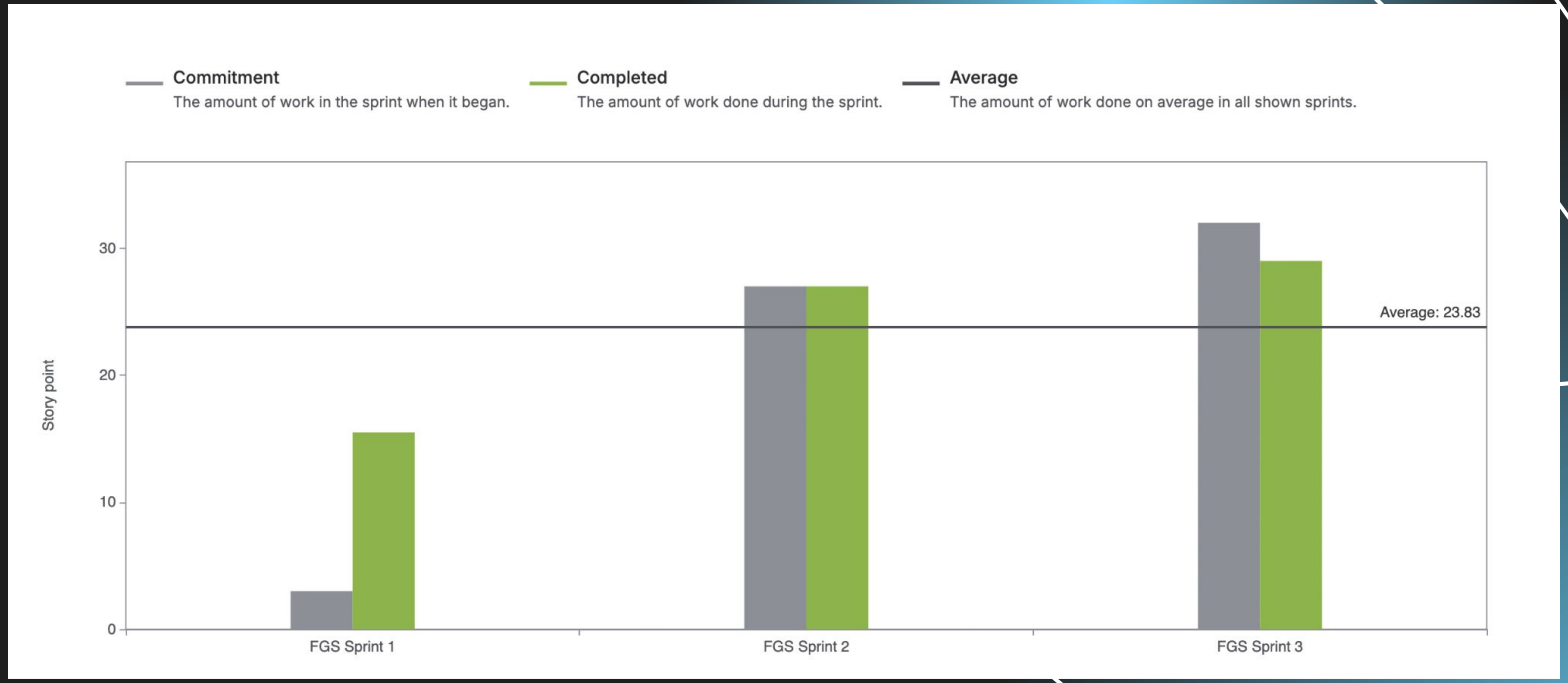
To Do

- Integrate Kit ID service with backup and FHIR export
- Add central UUID validation and registration

Burndown Chart



Velocity Chart



Retrospective

▶️ Start doing	🛑 Stop doing	🙌 Keep doing
Weekly Standups	Not having a fixed team meeting time	Log actual hours in Jira
Plan Sprint tasks and goals in advance	Not updating each other regularly on progress	Progress Update in Jira
Pushing changes regularly to GitHub		

Next steps

Step 1

Add central kitID validation and registration.

Step 2

Begin implementing licensing handshake between organizations and admin, integrate sbt toolchain, and encryption into kit for all data transfer.

Step 3

Thank you!

Questions?